How BARCS Finds Important Genes

Chapter 2: The calculations

How BARCS fits the model and decides if an effect is real

Prepared for Hyun-Hwan Jeong and Kyu-Won Lee

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About this guide. This guide explains how the BARCS computer program works. You need only arithmetic, fractions, percents, powers and square roots. The guide teaches logarithms when it uses them. Both chapters use the same small set of numbers, so you can follow every step. The guide uses the writing rules of ASD-STE100 Simplified Technical English.

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What you need from Chapter 1

This chapter continues the example of Chapter 1. Here is a short summary.

The data. One guide, four libraries:

Library	Group	Guide count K	Total count N
1	control	180	1,000,000
2	control	240	1,200,000
3	drug	120	1,000,000
4	drug	80	800,000

- **Proportion** $p = K/N$. Combined: 191 per million (control) and 111 per million (drug). The fold change is 0.58.
- **Log-odds** $= \ln\Big(p/(1-p)\Big)$: -8.56 (control) and -9.10 (drug).
- **Noise.** The variance of a count is $N \times \mu \times (1-\mu) \times \Big(1 + (N-1) \times \rho\Big)$. Here μ is the expected proportion. ρ is the correlation between two reads from the same library; it measures the biological noise. The wobble of the true proportion between replicates is $V = \rho \times \mu \times (1-\mu)$.
- **Two rules.** Variances of independent values add up. Multiplying a value by c multiplies its variance by $c \times c$.
- **Design matrix** X: one row for each library, one column for each β. Our X has the rows $(1,0)$, $(1,0)$, $(1,1)$, $(1,1)$.
- **Prediction.** Multiply a row of X by the β values and add. The result is the predicted log-odds η. Then $\mu = e^\eta/(1+e^\eta)$.
- β_0 is the control log-odds. β_1 is the effect of the drug.

Step 6: How BARCS finds the best β values

BARCS cannot find β_0 and β_1 in one calculation. It uses a loop of "guess, check, improve". The name of this method is **iteratively reweighted least squares** (IRLS). Each pass through the loop is an **iteration**.

The loop

1. **Guess.** Start with the simple log-odds of each group (Step 3 in Chapter 1).
2. **Predict.** Use the design matrix to predict η and μ for each library (Step 5 in Chapter 1).

3. **Measure the noise.** Find ρ from the differences between the real and the expected counts (Step 7).
4. **Weigh.** Give each library a **weight**. A library with less noise gets a larger weight.
5. **Correct.** Calculate a corrected value, the **working value**, for each library.
6. **Improve.** Take a weighted average of the working values in each group. These averages give the new β values.
7. **Repeat** from part 2 until the β values stop changing.

The working value: a straight-line step

BARCS works on the log-odds scale. The real log-odds of library i is $\ln\Big((K_i/N_i)/(1 - K_i/N_i)\Big)$. But BARCS does not use this value directly. If $K_i = 0$, the value is $\ln(0)$, which does not exist. So BARCS uses a straight line instead.

The slope of the log-odds. Near a proportion μ, the log-odds changes by about $1 \div \Big(\mu \times (1 - \mu)\Big)$ for each unit of change in the proportion. Here is why. The log-odds is $\ln(p) - \ln(1 - p)$. Look at the two parts one at a time, with $p = 0.2$ and a small step of 0.001:

Part	At $p = 0.2$	At $p = 0.201$	Change $\div$ 0.001
$\ln(p)$	-1.60944	-1.60445	$4.99 \approx 1 \div 0.2 = 5$
$-\ln(1 - p)$	$+0.22314$	$+0.22439$	$1.25 = 1 \div 0.8$
log-odds (sum)	-1.38629	-1.38006	$6.24 \approx 6.25$

So the slope is $1 \div p + 1 \div (1 - p)$. Put the two fractions over one bottom:

$$\frac{1}{p} + \frac{1}{1-p} = \frac{(1-p)+p}{p \times (1-p)} = \frac{1}{p \times (1-p)}.$$

Check: $1 \div (0.2 \times 0.8) = 1 \div 0.16 = 6.25$.

The working value. Start at the current prediction η_i (the log-odds of μ_i). Move along the straight line by the distance $K_i/N_i - \mu_i$:

$$v_i = \eta_i + (K_i/N_i - \mu_i) \times \frac{1}{\mu_i \times (1 - \mu_i)}.$$

The letter v stands for "working value". It is close to the real log-odds, and it exists even when $K_i = 0$.

Worked example 10a: working value of library 1

Use the first guess $\mu_1 = 0.000190909$, so $\eta_1 = -8.563523$.

- Distance: $K_1/N_1 - \mu_1 = 0.000180 - 0.000190909 = -0.0000109$.
- Slope: $1 \div (0.000190909 \times 0.999809) = 1 \div 0.000190873 = 5{,}239$.
- Step: $-0.0000109 \times 5{,}239 = -0.0572$.
- Working value: $v_1 = -8.5635 + (-0.0572) = -8.6207$.

The real log-odds is $\ln(0.000180/0.999820) = -8.6224$. The straight line gives -8.6207. They are very close.

The weight: one divided by the variance

Why weights? Suppose you have two measurements of the same thing. One has the variance 1. The other has the variance 4 (it is noisier).

- With equal weights, the average has the variance $(1 + 4) \div (2 \times 2) = 1.25$.
- With the weights $1 \div 1 = 1$ and $1 \div 4 = 0.25$, the average has the variance $1 \div (1 + 0.25) = 0.8$. That is smaller.

The rule "weight = $1 \div$ variance" gives the average with the smallest variance. So BARCS gives each working value the weight

$$w_i = \frac{1}{\text{variance of } v_i}.$$

The variance of the working value. We build it from Chapter 1 in three steps. The right column uses library 1 in iteration 1 ($N = 1{,}000{,}000$, $\mu = 0.000190909$, $\rho = 0.0000003841$).

Quantity	**Formula**	**Library 1**
1. variance of K_i	$N_i \times \mu_i \times (1 - \mu_i) \times \big(1 + (N_i - 1) \times \rho\big)$	$190.87 \times 1.3841 = 264.19$
2. variance of K_i/N_i	line 1 $\div (N_i \times N_i)$	$264.19 \div 10^{12}$
		$= 2.642 \times 10^{-10}$
3. variance of v_i	line 2 $\div \big(\mu_i \times (1 - \mu_i)\big)^2$	$2.642 \times 10^{-10} \div 0.000190873^2$
		$= 0.007251$
4. weight w_i	$1 \div$ line 3	$1 \div 0.007251 = 137.90$

- **Line 1** is the variance formula of Chapter 1, Step 4.

- **Line 2:** dividing a value by N divides its variance by $N \times N$ (Chapter 1, Part 4 used the same rule with multiplying).
- **Line 3:** the working value multiplies $K_i/N_i - \mu_i$ by the slope $1 \div \Big(\mu_i \times (1-\mu_i)\Big)$. So the variance is multiplied by the slope squared. Adding η_i does not change the variance.
- **Line 4:** put lines 1–3 together and simplify. One N_i and one $\mu_i \times (1-\mu_i)$ cancel:

$$w_i = \frac{1}{\text{variance of } v_i} = \frac{N_i \times N_i \times \Big(\mu_i \times (1-\mu_i)\Big)^2}{N_i \times \mu_i \times (1-\mu_i) \times \Big(1 + (N_i - 1) \times \rho\Big)} = \frac{N_i \times \mu_i \times (1-\mu_i)}{1 + (N_i - 1) \times \rho}.$$

Check: $190.87 \div 1.3841 = 137.90$, the same as line 4.

A library with more reads gets a bigger top, so a bigger weight. More biological noise (ρ) gives a bigger bottom, so a smaller weight.

The new β values

For our design matrix, the new β values are weighted averages of the working values:

$$\beta_0 = \frac{w_1 \times v_1 + w_2 \times v_2}{w_1 + w_2}, \qquad \beta_0 + \beta_1 = \frac{w_3 \times v_3 + w_4 \times v_4}{w_3 + w_4}.$$

(For other design matrices, BARCS solves a small set of equations instead. The idea is the same.)

Why a loop? The weights and the working values use μ. But μ comes from the β values, and the β values come from the weights. So BARCS starts with a guess and repeats until nothing changes.

Iteration 1 in full

Worked example 10b: iteration 1

Guess. From Step 3 in Chapter 1, the control log-odds is -8.563522 and the drug log-odds is -9.104869. So $\beta_0 = -8.563522$ and $\beta_1 = -9.104869 - (-8.563522) = -0.541347$.

Predict. The control libraries get $\mu = 0.000190909$. The drug libraries get $\mu = 0.000111111$.

Measure the noise. Step 7 gives $\rho = 0.0000003841$ for these predictions.

Weigh library 1. Top: $1{,}000{,}000 \times 0.000190909 \times 0.999809 = 190.87$.
Bottom: $1 + 999{,}999 \times 0.0000003841 = 1.3841$.
Weight: $w_1 = 190.87 \div 1.3841 = 137.90$.

Correct library 1. Example 10a: $v_1 = -8.6207$.

The same calculations for all four libraries give:

Library	Weight w_i	Working value v_i	$w_i \times v_i$
1 (control)	137.905	−8.62068	−1188.84
2 (control)	156.785	−8.51589	−1335.16
3 (drug)	80.269	−9.02486	−724.42
4 (drug)	67.988	−9.20488	−625.82

Improve.
Control: $\beta_0 = (-1188.84 - 1335.16) \div (137.905 + 156.785) = -2524.00 \div 294.69 = -8.564929$.
Drug: $\beta_0 + \beta_1 = (-724.42 - 625.82) \div (80.269 + 67.988) = -1350.24 \div 148.257 = -9.107414$.
So $\beta_1 = -9.107414 - (-8.564929) = -0.542486$.

The next iterations

BARCS repeats the loop with the new values. The changes become very small:

After iteration	β_0	β_1
0 (the guess)	−8.563522	−0.541347
1	−8.564929	−0.542486
2	−8.564934	−0.542492
3	−8.564934	−0.542492

After iteration 3, the values do not change in the first six decimal places. BARCS stops. The final values are $\beta_0 = -8.565$ and $\beta_1 = -0.542$.

Remember

The weighted average gives a slightly different answer from the simple log-odds of Step 3 in Chapter 1 (-0.5425 instead of -0.5413). The difference comes from the weights. Library 2 has the most reads and the least relative noise. So it counts the most.

Step 7: How BARCS measures the biological noise

Step 4 in Chapter 1 used $\rho = 0.000000386$. This section shows where the number comes from.

Pearson residuals

For each library, BARCS subtracts the expected count from the real count. It divides the result by the standard deviation from Step 4 in Chapter 1. The result is a **Pearson residual**:

$$\text{Pearson residual}_i = \frac{K_i - N_i \times \mu_i}{\sqrt{N_i \times \mu_i \times (1 - \mu_i) \times \Big(1 + (N_i - 1) \times \rho\Big)}}.$$

A Pearson residual tells how many standard deviations a library is from its prediction.

The rule for ρ

Why the sum should be the degrees of freedom. A Pearson residual is a distance measured in standard deviations. If the noise model is correct, its square is 1 on average: (squared distance) ÷ (variance) = variance ÷ variance = 1. So with four libraries, the four squares add up to about 4.

But BARCS does not know the true μ. It uses the fitted μ, which the β values pull toward the data. The fit makes the residuals a little smaller. Each fitted β value takes away 1 from the sum, on average. So the expected sum is

$$(\text{number of libraries}) - (\text{number of } \beta \text{ values}) = 4 - 2 = 2 = \text{df}.$$

A computer checked this. It made 20,000 artificial guides with the true ρ and four libraries like ours. With the true μ, the sum of the squared Pearson residuals was 4.00 on average. With the fitted μ, it was 2.00 on average.

The rule. BARCS turns this around. It chooses ρ so that the sum is exactly the degrees of freedom:

$$\text{sum of the squared Pearson residuals} = \text{df}.$$

(Statisticians call this a *method of moments*: make a sum that you can calculate equal to the value it should have on average.)

Worked example 11: start with $\rho = 0$

Use the expected counts from Step 5 in Chapter 1. With $\rho = 0$, the bottom of each residual is the counting noise only.

Library	$(K - N \times \mu)^2$	$N \times \mu \times (1 - \mu)$	Squared residual
1	$(-10.64)^2 = 113.21$	190.60	0.594
2	$(+11.23)^2 = 126.16$	228.72	0.552
3	$(+9.17)^2 = 84.14$	110.82	0.759
4	$(-8.66)^2 = 75.03$	88.65	0.846
		Sum	2.751

The sum is 2.751. The degrees of freedom are 2. The sum is too big. So there is more noise than counting noise alone. We need $\rho > 0$.

Worked example 12: guess and check for ρ

A larger ρ makes the bottom of each residual larger. So the sum goes down. BARCS tries values until the sum equals 2:

ρ	Sum of squared residuals
0	2.751
0.0000001	2.506
0.0000002	2.302
0.0000003	2.129
0.000000386	2.000
0.0000005	1.851

The answer is $\rho = 0.000000386$. If the sum is already 2 or less at $\rho = 0$, BARCS uses $\rho = 0$.

The ratio $2.751 \div 2 = 1.376$ is also useful. It is the **variance inflation**. At $\rho = 0$, the sum should be about 2; it is 2.751. So the variance is about 1.38 times bigger than counting noise alone. This is close to the extra factor $1 + (N - 1) \times \rho = 1.386$ of Chapter 1. Step 9 uses the inflation.

Step 8: Is the effect real?

The effect is $\beta_1 = -0.542$. But replicates are noisy. Maybe the true effect is 0 and the noise made the drop. BARCS tests this question in four parts.

The standard error

The **standard error** (SE) tells how much the estimate of β_1 can change because of noise. A small SE means a precise estimate.

For our design matrix, the standard error comes from the weights of Step 6. Add the weights of each group:

$$W_{\text{control}} = w_1 + w_2, \qquad W_{\text{drug}} = w_3 + w_4.$$

Then

$$\text{SE}(\beta_1) = \sqrt{\frac{1}{W_{\text{control}}} + \frac{1}{W_{\text{drug}}}}.$$

Where this formula comes from. It uses only three facts: the weight is $1 \div$ variance (Step 6); variances of independent values add up; and multiplying a value by c multiplies its variance by $c \times c$.

Step	**Why**
1. variance of $v_i = 1 \div w_i$	definition of the weight
2. $\beta_0 = \dfrac{w_1}{W_c} \times v_1 + \dfrac{w_2}{W_c} \times v_2 \quad (W_c = w_1 + w_2)$	Step 6
3. variance of $\beta_0 = \left(\dfrac{w_1}{W_c}\right)^2 \times \dfrac{1}{w_1} + \left(\dfrac{w_2}{W_c}\right)^2 \times \dfrac{1}{w_2}$	multiply by $c \Rightarrow$ variance $\times\, c \times c$; add
4. $= \dfrac{w_1 + w_2}{W_c \times W_c} = \dfrac{1}{W_c}$	one w cancels in each part
5. variance of $(\beta_0 + \beta_1) = 1 \div W_{\text{drug}}$	the same for the drug libraries
6. variance of $\beta_1 = \dfrac{1}{W_c} + \dfrac{1}{W_{\text{drug}}}$	$\beta_1 = (\beta_0 + \beta_1) - \beta_0$; variances add
7. $\text{SE}(\beta_1) = \sqrt{\text{line 6}}$	SE is the standard deviation of the estimate

Line 6 adds the variances even though it subtracts the values. The reason: $-1 \times -1 = +1$, so a minus sign does not change a variance.

More weight means less noise, so the SE becomes smaller. (For other design matrices, BARCS uses the inverse of the matrix $X^\top W X$. For our design, that calculation gives exactly this formula.)

Worked example 13: the standard error

The final weights are 137.524, 156.322, 79.955 and 67.737.
$W_{\text{control}} = 137.524 + 156.322 = 293.846$.
$W_{\text{drug}} = 79.955 + 67.737 = 147.692$.
$1 \div 293.846 = 0.003403$ (line 4: $(137.524 + 156.322) \div 293.846^2$ gives the same).
$1 \div 147.692 = 0.006771$.
Sum: $0.003403 + 0.006771 = 0.010174$.
$\text{SE} = \sqrt{0.010174} = 0.101$.
The control group has about twice the weight of the drug group. So most of the uncertainty comes from the drug group.

The t value

The **t value** compares the effect with its standard error:

$$t = \frac{\beta_1}{\text{SE}}.$$

A t value far from 0 means that the effect is large compared with the noise.

Degrees of freedom

The **degrees of freedom** (df) tell how much information we have about the noise:

$$\text{df} = (\text{rows of } X) - (\text{columns of } X) = (\text{number of libraries}) - (\text{number of } \beta \text{ values}).$$

Fewer degrees of freedom mean less information. Then BARCS needs a bigger t value before it accepts an effect as real. (The reason is that the SE itself is estimated from few libraries, so it can be too small by chance. W. S. Gosset, who wrote as "Student", worked out the table for this case in 1908.[1])

Degrees of freedom	1	2	14	very many
$\lvert t \rvert$ needed for $p < 0.05$	12.71	4.30	2.14	1.96

The p-value

The **p-value** answers this question: "If the true effect is 0, how often does noise alone give a t value this far from 0?" A small p-value means that noise alone seldom gives such a result. Scientists often use 0.05 as a limit.

[1] Student (1908). The probable error of a mean. *Biometrika* 6: 1–25.

Worked example 14: testing the effect

1. $t = -0.542 \div 0.101 = -5.38$.
2. X has 4 rows and 2 columns. So df $= 4 - 2 = 2$.
3. The table shows that 2 degrees of freedom need $|t| > 4.30$. Our $|t| = 5.38$. It is larger than 4.30. So $p < 0.05$.
4. A computer gives the exact p-value: $p = 0.033$.

The p-value is less than 0.05. So the drop is probably not only noise.

Remember

The degrees of freedom come from the number of libraries, not from the number of reads. A library with 100 million reads still counts as one row of X.

Step 9: Borrow information from other guides

With only 2 degrees of freedom, the noise estimate of one guide is not reliable. But a screen has thousands of guides. Most guides have a similar amount of noise. So BARCS combines two estimates:

- the variance inflation of this guide (Step 7), and
- the typical variance inflation of guides with similar counts.

This method is **moderation**.[2] BARCS takes a weighted average. Each estimate gets a number of "votes". The guide's own estimate gets its degrees of freedom as votes. The typical estimate gets the **prior degrees of freedom**, d_0. BARCS calculates d_0 and the typical value from all the guides in the screen.

$$\text{moderated inflation} = \frac{\text{df} \times \text{own inflation} + d_0 \times \text{typical inflation}}{\text{df} + d_0}.$$

[2]BARCS uses the empirical-Bayes method of G. K. Smyth (2004). Linear models and empirical Bayes methods for assessing differential expression in microarray experiments. *Statistical Applications in Genetics and Molecular Biology* 3: Article 3.

Worked example 15: moderation with simple numbers

A guide has 3 degrees of freedom. Its own inflation is 4.0. The typical inflation is 1.5, with $d_0 = 12$ votes.

$$\text{moderated inflation} = \frac{3 \times 4.0 + 12 \times 1.5}{3 + 12} = \frac{12 + 18}{15} = \frac{30}{15} = 2.0.$$

The guide's value moves from 4.0 toward 1.5. It stops at 2.0.

What moderation does to the test

BARCS uses the moderated inflation in three places:

1. It multiplies the standard error by $\sqrt{\text{moderated inflation} \div \text{own inflation}}$. Here is why. The variance of β_1 grows in proportion to the inflation. To swap the own inflation for the moderated one, multiply the variance by (moderated ÷ own). The SE is the square root of the variance, so it is multiplied by the square root. (If the own inflation is below 1, BARCS uses 1, so moderation never makes a test less careful than counting noise alone.)
2. It recalculates t with the new standard error.
3. It adds the d_0 votes to the degrees of freedom. So the test uses $\text{df} + d_0$.

Worked example 16: moderation for our guide

Our guide has own inflation 1.376 (Step 7) and 2 degrees of freedom. Suppose that the screen gives typical inflation 1.5 with $d_0 = 12$.
Moderated inflation: $(2 \times 1.376 + 12 \times 1.5) \div (2 + 12) = (2.751 + 18) \div 14 = 1.482$.
Factor for the SE: $\sqrt{1.482 \div 1.376} = \sqrt{1.077} = 1.038$.
New SE: $0.101 \times 1.038 = 0.105$.
New t: $-0.542 \div 0.105 = -5.18$.
New degrees of freedom: $2 + 12 = 14$.
With 14 degrees of freedom, $|t|$ must be larger than 2.14 for $p < 0.05$. Our $|t| = 5.18$.
The new p-value is 0.00014.
The SE became slightly larger, because the typical guide is a little noisier than our guide. But the degrees of freedom grew from 2 to 14. So the test became much more powerful.

BARCS uses moderation when it has 50 or more usable guides.

Step 10: Check the results with control guides

A good screen contains **non-targeting control guides**. These guides switch off no gene. So their true effect is 0.

If the noise model is correct, about 5% of control guides get $p < 0.05$ by chance. If many more control guides get $p < 0.05$, the noise model is too small. Then BARCS makes all the tests wider.

Worked example 17: calibration with control guides

A screen has 100 control guides. We expect about 5 of them to get $p < 0.05$. But 15 of them get $p < 0.05$. The noise is too small.
BARCS finds a **scale factor** that makes the control guides behave correctly. It sorts the 100 control guides by the size of $|t|$ (the value without its sign). It takes the value that 95 of the 100 guides stay below. Suppose that this value is 2.47. With many degrees of freedom, the t table says that this value should be 1.96. So the scale factor is $2.47 \div 1.96 = 1.26$. BARCS divides every t value by 1.26.
A gene with $t = 3.5$ now gets $t = 3.5 \div 1.26 = 2.78$. Its p-value becomes larger. This change protects you from false results.

The scale factor is never less than 1. So calibration can only make the results more careful.

Step 11: From guides to genes

Each gene usually has several guides, often 3 to 6. Each guide gives its own result. BARCS combines them into one result for the gene. It uses the **Stouffer method** in three parts.

1. Change each guide's p-value into a **z score**. A z score tells how many standard deviations a result is from 0. The sign of the z score follows the sign of the effect.
2. Add the z scores.
3. Divide the sum by the square root of the number of guides, m.

$$Z_{\text{gene}} = \frac{z_1 + z_2 + \cdots + z_m}{\sqrt{m}}.$$

Why divide by $\sqrt{m}$? If the gene has no effect, each z score has the variance 1. The guides are independent, so the variance of the sum is $1 + 1 + \cdots + 1 = m$ (Chapter 1, Part 2: variances add). Dividing a value by $\sqrt{m}$ divides its variance by $\sqrt{m} \times \sqrt{m} = m$. So Z_{gene} has the variance $m \div m = 1$ again, and the same z table works for it.[3]

To change a p-value into a z score, use the table of the normal distribution. A two-sided p-value of 0.05 belongs to $z = 1.96$. Some other pairs:

Two-sided p-value	0.317	0.134	0.046	0.012	0.0027
z score	1.0	1.5	2.0	2.5	3.0

[3] S. A. Stouffer and others (1949). *The American Soldier*, Vol. 1. Princeton University Press.

If the effect is negative (the guide drops), the z score gets a minus sign.

Worked example 18: combining three guides

A gene has $m = 3$ guides. Their p-values are 0.046, 0.134 and 0.012. Their z scores are 2.0, 1.5 and 2.5.

$$Z_{\text{gene}} = \frac{2.0 + 1.5 + 2.5}{\sqrt{3}} = \frac{6.0}{1.732} = 3.46.$$

This Z gives $p = 0.00053$ for the gene. No single guide is that strong. Together, the three guides give strong evidence.

When guides share noise

Sometimes the guides of one gene share noise. Then they agree more than they should by chance. BARCS measures this sharing with the **guide correlation**, r. Then the variance of the sum is not m. Use the squares-and-pairs rule from Chapter 1, Part 4:

- m squares, each with the variance 1: together m;
- $m \times (m - 1)$ pairs, each with the covariance r: together $m \times (m - 1) \times r$.

So BARCS divides by the square root of this larger variance:

$$Z_{\text{gene}} = \frac{z_1 + z_2 + \cdots + z_m}{\sqrt{m + m \times (m - 1) \times r}}.$$

Worked example 19: shared noise

Use the same three guides and $r = 0.1$. The bottom becomes $\sqrt{3 + 3 \times 2 \times 0.1} = \sqrt{3.6} = 1.897$. So $Z_{\text{gene}} = 6.0 \div 1.897 = 3.16$, and $p = 0.0016$. The evidence is still strong, but less strong than before.

Step 12: Many genes and false discoveries

A screen tests about 20,000 genes. Suppose that no gene has an effect. If we use the limit $p < 0.05$, about 5% of the genes still pass by chance. That is about 1,000 false results.

So BARCS uses the **false discovery rate** (FDR). The FDR is the expected fraction of false results in your list of hits.

The idea. When a gene has no effect, its p-value is equally likely to be anywhere from 0 to 1. So a fraction p of such genes get a p-value of p or less by chance. Example: of 20,000

genes with no effect, about $20{,}000 \times 0.001 = 20$ get $p \leq 0.001$. Now suppose that your list "all genes with $p \leq 0.001$" has 100 genes. At most about 20 of them are there by chance. So the false fraction is about $20 \div 100 = 0.2$. In general:

$$\text{FDR of a list} \approx \frac{p \times (\text{number of genes})}{\text{number of genes in the list}}.$$

If you sort the genes by p-value, the list that ends at a gene has "rank" genes in it. That gives part 3 below.[4]

BARCS calculates it with the **Benjamini–Hochberg method**:

1. Sort the genes from the smallest p-value to the largest.
2. Give each gene a rank: 1, 2, 3, and so on.
3. For each gene, calculate $p \times (\text{number of genes}) \div \text{rank}$.
4. Go from the bottom of the list to the top. If a value is larger than the value below it, replace it with the value below it. (A gene is in every longer list too. If a longer list has a smaller FDR, the gene can use that list.)

Worked example 20: false discovery rate for five genes

Gene	Rank	p	$p \times 5 \div \text{rank}$	FDR (after part 4)
A	1	0.001	$0.001 \times 5 \div 1 = 0.005$	0.005
B	2	0.008	$0.008 \times 5 \div 2 = 0.020$	0.020
C	3	0.039	$0.039 \times 5 \div 3 = 0.065$	0.051 (from D)
D	4	0.041	$0.041 \times 5 \div 4 = 0.051$	0.051
E	5	0.600	$0.600 \times 5 \div 5 = 0.600$	0.600

With the limit FDR < 0.05, genes A and B are hits. With the limit FDR < 0.10, genes A, B, C and D are hits.

Step 13: Let the control guides decide (new in BARCS 0.2.2)

Step 10 checks the noise model with the control guides and makes all t values smaller by one scale factor. Sometimes that is not enough. In some screens, the control guides have *more extreme* values than any scaling can fix. This happens, for example, when the replicate flasks were split from one population of cells and share the same large clones. Then the noise model is wrong in its shape, not only in its size.

[4] Y. Benjamini and Y. Hochberg (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B* 57: 289–300.

BARCS 0.2.2 has an answer for this case: the function `bb_gene_empirical_null()`. Its idea is simple: **do not trust the model for "no effect". Ask the control guides what "no effect" looks like in this experiment.**

Pseudo-genes

A control guide has no true effect. So a "gene" made of control guides shows what a gene with no effect looks like. BARCS builds many of these **pseudo-genes**:

1. Take a real gene with m guides.
2. Pick m control guides at random. This is one pseudo-gene.
3. Summarize it in the same way as a real gene: the average of the signed z scores of its guides.
4. Repeat many times, for example 999 times.

The 999 pseudo-gene values show the spread of "no effect" in your own data, with all its real noise.

The p-value from a count

Now compare the real gene with the pseudo-genes. Count how many pseudo-genes are at least as extreme. The p-value is

$$p = 2 \times \frac{1 + (\text{number of pseudo-genes at least as far out on the same side})}{1 + (\text{number of pseudo-genes})}.$$

The "1+" counts the real gene itself, so p is never 0. The "2×" makes the test two-sided, as in Step 8.

Worked example 21: a p-value from pseudo-genes

A gene has 4 guides. The average of their z scores is +2.4. BARCS draws 999 pseudo-genes of 4 control guides each. Only 4 of them have an average of +2.4 or more.

$$p = 2 \times \frac{1 + 4}{1 + 999} = 2 \times \frac{5}{1{,}000} = 0.01.$$

Then BARCS uses the Benjamini–Hochberg method of Step 12 on these p-values.

Widen the pseudo-genes for shared noise

The guides of a real gene can share noise (the guide correlation r of Step 11). Random control guides do not share noise, because they come from different places. So a pseudo-gene would

wobble less than a real gene with no effect. BARCS fixes this. It moves each pseudo-gene away from the centre by the factor $\sqrt{1 + (m - 1) \times r}$.

This factor comes from the squares-and-pairs rule of Chapter 1. The average of m z scores with correlation r has the variance

$$\frac{m + m \times (m - 1) \times r}{m \times m} = \frac{1}{m} \times \Big(1 + (m - 1) \times r\Big).$$

Without correlation it is $1 \div m$. So the variance is $1 + (m - 1) \times r$ times bigger, and the standard deviation is $\sqrt{1 + (m - 1) \times r}$ times bigger. For $m = 4$ and $r = 0.2$: $\sqrt{1 + 3 \times 0.2} = \sqrt{1.6} = 1.26$.

Limits

- **The smallest p-value.** With n pseudo-genes (or n control guides for genes of one guide), the smallest possible p-value is $2 \div (n + 1)$. With 1,000 controls, it is $2 \div 1{,}001 = 0.002$. Benjamini–Hochberg over 20,000 genes then needs about $0.002 \times 20{,}000 \div 0.1 = 400$ genes at that floor before it calls any gene at an FDR of 0.10. So "no hits" from the empirical null can mean "the controls are too few to confirm the hits", not "there are no hits".
- **Offset controls.** Non-targeting control guides make no cut in the DNA. In knockout screens, every targeting guide pays a small cost for the cut, so all targeting guides sit a little away from the controls. The option `centre = "targets"` takes the centre of "no effect" from the targeting guides and only the spread from the controls.
- **It is optional.** BARCS uses it only when you ask, and it needs at least 100 usable control guides. The model results stay available next to it.

Remember

The model says what "no effect" should look like. The control guides show what it really looks like in your experiment. When the two disagree, the empirical null trusts the controls.

Two special cases

A guide with zero reads

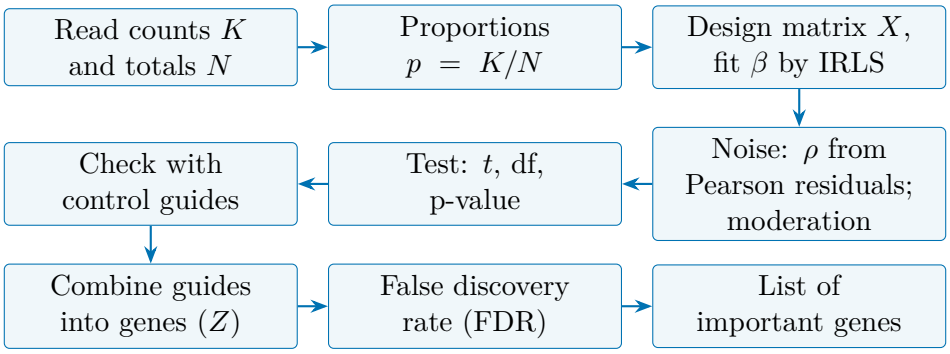
Sometimes a guide has 0 reads in all control libraries. Then the log-odds of the control group is $\ln(0)$. This value does not exist, because no power of e equals 0. Older versions of BARCS could not fit such a guide. So they removed it.

BARCS version 0.2.1 adds a very small number to the counts first. This number is less than one read. The method is the **Firth correction**. Now the guide gets a value, and BARCS keeps it.

A very large change

Sometimes a guide grows by hundreds of times. The usual test from Step 8 can then fail. The standard error grows faster than the effect, so the t value becomes small. BARCS version 0.2.1 has a second test, the **likelihood-ratio test**. It compares how well the model fits with the effect and without the effect. The option `test = "auto"` uses this test only for guides with very large changes.

Summary: the full path



All symbols in one table

Symbol	Meaning	Value in our example
K	read count of the guide in one library	180, 240, 120, 80
N	total read count of one library	1.0, 1.2, 1.0, 0.8 million
p	proportion, K/N	191 and 111 per million
fold change	$p_{\text{drug}} \div p_{\text{control}}$	0.58
$\log_2$	number of doublings	$\log_2(0.58) = -0.78$
ln	natural logarithm (powers of $e = 2.718$)	$\ln(0.58) = -0.54$
log-odds	$\ln(p/(1-p))$	-8.56 (control)
μ	expected proportion	0.000191 (control)
ρ	correlation of two reads from one library (biological noise)	0.000000386
X	design matrix: a row per library, a column per β	4 rows, 2 columns
β_0	control log-odds in the model	-8.565
β_1	effect of the drug	-0.542
η	predicted log-odds of a library (row of X times β)	-8.565, -9.107
w	weight of a library in the fit	137.5 (library 1)
working value	prediction plus a straight-line step (Example 10a)	-8.621 (library 1)
Pearson residual	distance from the prediction in SDs	-0.65 (library 1)
inflation	noise compared with counting noise	1.376
SE	standard error of the effect	0.101
t	effect divided by SE	-5.38
df	libraries minus β values	$4 - 2 = 2$
p-value	chance of this result if the effect is 0	0.033
d_0	prior degrees of freedom (votes)	12 (Example 15)
z, Z	z score of a guide, of a gene	3.46 (Example 18)
m	number of guides for a gene	3
r	guide correlation (shared noise)	0.1 (Example 19)
FDR	expected fraction of false hits	0.005 for gene A
pseudo-gene	m random control guides, summarized like a gene	$p = 0.01$ (Example 21)

Note: the letter p has two uses in this guide. Alone, p is a proportion. In the word "p-value", it is a probability. The text always tells you which one it is.

Exercises for Chapter 2

These exercises continue the screen of the Chapter 1 exercises (guide g7). Work on paper and show each calculation. The answers are in a separate answer sheet.

The tags tell you what to do: [Calculate] work out a number, [Explain] say why in your own words, [Decide] choose and give a reason.

Values to use. Expected proportions: $\mu = 0.00015$ (control) and $\mu = 0.000075$ (drug). Biological noise: $\rho = 0.000001$. Effect: $\beta_1 = -0.69$. Use ln for the natural logarithm.

Part A: Fitting

1. [Calculate] [Explain] A control library has $K/N = 0.00018$. The current prediction is $\mu = 0.00015$, so $\eta = \ln(0.00015/0.99985) = -8.8047$.

(a) Calculate the slope $1 \div \big(\mu \times (1 - \mu)\big)$.

(b) Calculate the step $(K/N - \mu) \times$ slope and the working value $v = \eta +$ step.

(c) Calculate the real log-odds $\ln(0.00018/0.99982)$ and compare.

(d) Why does BARCS use the straight-line working value and not the real log-odds? Think of a guide with $K = 0$.

2. [Calculate] [Explain] Two control libraries have $N = 1{,}000{,}000$ and $N = 2{,}000{,}000$. Use $\mu = 0.00015$.

(a) Calculate the weight $w = N \times \mu \times (1 - \mu) \div \big(1 + (N - 1) \times \rho\big)$ of each library.

(b) The second library has twice the reads. Does it have twice the weight? Explain why not.

(c) What would the two weights be if $\rho = 0$?

3. [Calculate] [Explain] Two libraries have the working values -8.60 (weight 75) and -8.90 (weight 100).

(a) Calculate the weighted average.

(b) Calculate its standard error, $\sqrt{1 \div (75 + 100)}$.

(c) The weighted average is closer to -8.90 than to -8.60. Why is that the right thing to do?

Part B: Is the effect real?

4. [Calculate] [Explain] The control weights add up to $W_{\text{control}} = 175$. The drug weights add up to $W_{\text{drug}} = 70$.

(a) Calculate $\mathrm{SE}(\beta_1) = \sqrt{1/W_{\text{control}} + 1/W_{\text{drug}}}$.

(b) Calculate $t = \beta_1 \div \mathrm{SE}$ and the degrees of freedom (4 libraries, 2 β values).

(c) With 2 degrees of freedom, $|t|$ must be larger than 4.30 for $p < 0.05$. Is the effect significant?

(d) Which group adds more uncertainty? What would you change in the next experiment to make the SE smaller?

5. [Calculate] [Explain] Two guides, P and Q, each have 4 libraries and 2 β values. At $\rho = 0$, the squared Pearson residuals add up to 5.1 for guide P and to 1.4 for guide Q.

(a) Calculate the variance inflation (sum ÷ df) of each guide.

(b) For which guide does BARCS use $\rho > 0$? For which does it use $\rho = 0$?

(c) Why do we compare the sum with 2 and not with 4?

6. [Calculate] [Explain] Guide P has its own inflation 2.55 with 2 degrees of freedom. In the screen, the typical inflation is 1.5 with $d_0 = 10$.

(a) Calculate the moderated inflation.

(b) Calculate the factor for the SE, $\sqrt{\text{moderated} \div \text{own}}$, and the new degrees of freedom.

(c) Why is it reasonable to borrow information from other guides? When could it be a bad idea?

Part C: From guides to a list of genes

7. [Calculate] [Decide] The screen has 300 non-targeting control guides. 45 of them get $p < 0.05$.

(a) How many would you expect if the noise model were correct? What does the real number tell you?

(b) 95% of the control guides have $|t|$ below 2.6. The t table says 1.96. Calculate the scale factor.

(c) A gene has $t = 3.0$. Calculate its new t. Is it still beyond 1.96?

8. [Calculate] [Explain] A gene has four guides with the z scores 2.2, 1.8, −0.3 and 2.5.

(a) Calculate Z_{gene} with the Stouffer method (no shared noise).

(b) Calculate Z_{gene} again with the guide correlation $r = 0.2$.

(c) Why does one guide with a negative z lower the result? Why does shared noise lower the result?

9. [Calculate] [Decide] A screen tests 10,000 genes. 40 genes have $p \leq 0.0005$.

(a) If no gene had an effect, about how many genes would have $p \leq 0.0005$ by chance?

(b) Estimate the false discovery rate of this list of 40 genes.

(c) You want a list with an FDR of about 0.05. Should you use a smaller or a larger p-value limit? Explain.

10. [Decide] [Explain] A collaborator says: "We will read 20 million reads per library. With so many reads, 2 replicates per group are enough." Use $\rho = 0.000001$ and the degrees of freedom to answer. (Hint: calculate the extra factor $1+(N-1)\times\rho$ for $N = 20{,}000{,}000$.)

11. [Calculate] [Explain] [Decide] (Step 13) In the control check of Exercise 7, the control guides still have too many extreme values after scaling. You try the empirical null.

(a) A gene has 3 guides with the z scores 2.0, 2.6 and 2.2. Calculate the average z score.

(b) BARCS draws 1,999 pseudo-genes of 3 control guides. 9 of them have an average of your answer to (a) or more. Calculate the p-value.

(c) The guide correlation is $r = 0.1$. By what factor does BARCS widen each pseudo-gene? Why is this needed?

(d) The screen has only 150 control guides and many genes with one guide. What is the smallest p-value a one-guide gene can get? What would you tell a student who sees "no hits" from the empirical null?